

Assignment - 1

Q.1
Ans. Given prices : 100, 108, 103, 115, 110

(a) Simple monthly returns

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Month 1, $R_1 = \frac{108 - 100}{100} = 8\%$

Month 2, $R_2 = \frac{103 - 108}{108} = -4.63\%$

Month 3, $R_3 = \frac{115 - 103}{103} = 11.65\%$

Month 4, $R_4 = \frac{110 - 115}{115} = -4.35\%$

(b) Log returns

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

$$r_1 = \ln\left(\frac{108}{100}\right) = 0.07696$$

$$r_2 = \ln\left(\frac{103}{108}\right) = -0.04740$$

$$r_3 = \ln\left(\frac{115}{103}\right) = 0.11020$$

$$r_4 = \ln\left(\frac{110}{115}\right) = -0.04445$$

(c) comparison of r_t and R_t

Month	Simple return R_t	Log return r_t	Difference
1	0.08000	0.07696	0.00304
2	-0.04630	-0.04740	0.00110
3	0.11650	0.11020	0.00630
4	-0.04348	-0.04445	0.00097

For small returns, $\ln(1+R) \approx R$

but approximation error $\uparrow$ as return magnitude $\uparrow$.

Q.2

Ans $\Rightarrow$ Given $E(R_1) = 15\%$ $E(R_2) = 7\%$
 $w_1 = 0.4$ $w_2 = 0.6$

(a) Linearity of expectation

$$E[aX + bY] = aE[X] + bE[Y]$$

Portfolio return: $R_p = w_1 R_1 + w_2 R_2$

$$\begin{aligned} E(R_p) &= w_1 E(R_1) + w_2 E(R_2) \\ &= 0.4(0.15) + 0.6(0.07) \\ &= 0.102 = 10.2\% \end{aligned}$$

(b) No, Expectation depends only on weighted averages of expected returns.

Correlation affects portfolio variance only.

Q.3

Ans $\Rightarrow$ $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

$$R_p = w_1 R_1 + w_2 R_2$$

$$\begin{aligned} \text{Var}(R_p) &= \text{Var}(w_1 R_1 + w_2 R_2) \\ &= w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2w_1 w_2 \text{Cov}(R_1, R_2) \end{aligned}$$

$$\text{Var}(R_1) = \sigma_1^2 \quad \text{Var}(R_2) = \sigma_2^2$$

$$\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$$

$$\Rightarrow \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2\rho_{12} \sigma_1 \sigma_2 w_1 w_2$$

Q.4

Ans $\Rightarrow$ $\mu_A = 18\%$ $\sigma_A = 25\%$

$\mu_B = 8\%$ $\sigma_B = 12\%$

$w_A = w_B = 0.5$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$$

Case I:

$$\begin{aligned}\sigma_p^2 &= (0.5)^2 (0.25)^2 + (0.5)^2 (0.12)^2 + 2(0.5)(0.5)(1)(0.25)(0.12) \\ &= 0.015625 + 0.0036 + 0.015 \\ &= 0.034225\end{aligned}$$

$$\sigma_p = \sqrt{0.034225} = 0.185 = 18.5\%$$

Case II: $\rho = 0$

$$\begin{aligned}\sigma_p^2 &= 0.015625 + 0.0036 \\ &= 0.019225\end{aligned}$$

$$\sigma_p = 13.87\%$$

Case III: $\rho = -1$

$$\begin{aligned}\sigma_p^2 &= 0.015625 + 0.0036 - 0.015 \\ &= 0.004225\end{aligned}$$

$$\sigma_p = 6.5\%$$

(b) weighted avg risk:

$$w_A \sigma_A + w_B \sigma_B$$

Squaring, $(w_A \sigma_A + w_B \sigma_B)^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B$

this equals to portfolio variance only when $\rho = 1$.

Hence, $\sigma_p = w_A \sigma_A + w_B \sigma_B$

only for perfectly positively correlated assets.

(c) For zero variance

$$w_A \sigma_A = w_B \sigma_B$$

since, $w_B = 1 - w_A$

$$w_A (0.25) = (1 - w_A) (0.12)$$

$$0.25 w_A = 0.12 - 0.12 w_A$$

$$w_A = \frac{0.12}{0.37} = 0.3243$$

$$w_A = 32.43\%$$

Q.5

Ans: When two assets have correlation less than 1, they do not move perfectly together. Therefore, negative movements in one asset may be partially offset by positive or smaller movements in the other.

In the portfolio variance formula,

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

the covariance term depends on correlation ρ . If $\rho < 1$, this term becomes smaller than the maximum possible value, reducing total portfolio variance. Hence, diversification lowers portfolio risk below the weighted average of individual risks. The lower the correlation, the greater the diversification benefit. When $\rho = -1$, risk can be eliminated completely for suitable weights.

Q.6
Ans → (a)

$$\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$$

Diagonal entries :

$$\begin{aligned}\sigma_1^2 &= 0.25^2 = 0.0625 \\ \sigma_2^2 &= 0.12^2 = 0.0144 \\ \sigma_3^2 &= 0.04^2 = 0.0016\end{aligned}$$

Off-diagonal entries :

$$\begin{aligned}\Sigma_{12} &= 0.4(0.25)(0.12) = 0.012 \\ \Sigma_{13} &= 0.1(0.25)(0.04) = 0.001 \\ \Sigma_{23} &= 0.2(0.12)(0.04) = 0.00096\end{aligned}$$

Thus,

$$\Sigma = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}$$

(b) A Covariance matrix is symmetric because :

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

Indeed

$$\Sigma_{12} = \Sigma_{21}$$

$$\Sigma_{13} = \Sigma_{31}$$

$$\Sigma_{23} = \Sigma_{32}$$

Hence, $\Sigma = \Sigma^T$

Q. 1
Ans =

$R_f = 4\%$

Sharpe ratio

$$S = \frac{E[R_p] - R_f}{\sigma_p}$$

(a) Portfolio X

$$S_X = \frac{0.14 - 0.04}{0.22} = 0.4545$$

Portfolio Y

$$S_Y = \frac{0.09 - 0.04}{0.10} = 0.50$$

Portfolio Z

$$S_Z = \frac{0.20 - 0.04}{0.35} = 0.4571$$

(b)

Portfolio

Sharpe ratio

Y

0.500

Z

0.457

X

0.455

A rational mean-variance investor prefers portfolio Y because it provides the highest excess return per unit of risk.

(c) Although Z has the highest absolute return, it also has much higher volatility. Portfolio Y delivers better risk-adjusted performance, meaning investors are compensated more efficiently for each unit of risk taken.

Q. 8

Ans (a) GMV portfolio variance
It is the no other

Portfolio $\Rightarrow$ the Global Minimum Variance (GMV) is the portfolio with the lowest possible variance among all feasible portfolios. It is the leftmost point on the efficient frontier because no other feasible portfolio has lower risk.

(b) No.

The lower half is called the inefficient frontier because investors can obtain higher expected return for the same level of risk by moving to the upper half.

Rational investors always choose portfolios on the efficient frontier.

- (C) The CML is constructed by drawing line from risk-free asset tangent to the efficient frontier. The tangency point is the market portfolio.
- $$\text{slope of CML} = \frac{E[R_M] - R_f}{\sigma_M}$$

which is slope ratio of market portfolio. It represents the best attainable risk-return tradeoff.

Q.9

Ans $\rightarrow$ (a) standard error:

$$SE(\hat{\mu}) = \frac{s}{\sqrt{n}} = \frac{0.30}{\sqrt{0.36}} = 0.05 = 5\%$$

This is very large relative to typical expected return implying historical mean estimates are ~~very~~ highly noisy and unreliable.

- (b) Mean-Variance Optimisation is extremely sensitive estimation errors in expected returns. Small errors in $\hat{\mu}$ can produce very large and unstable portfolio weights, because the optimiser aggressively allocates to assets with slightly higher estimated returns.

This is called error maximization.

Machine learning predicted returns help by incorporating more information and potentially generating more stable and accurate return forecasts, reducing estimation noise.

- (C) For N assets: $\frac{N(N+1)}{2}$

unique covariance parameters are needed.

for $N=50$

$$\frac{50(51)}{2} = 1275$$

estimating such a large number of parameters accurately is difficult because;

- large data requirements,
- estimation noise,
- instability,
- overfitting risk.